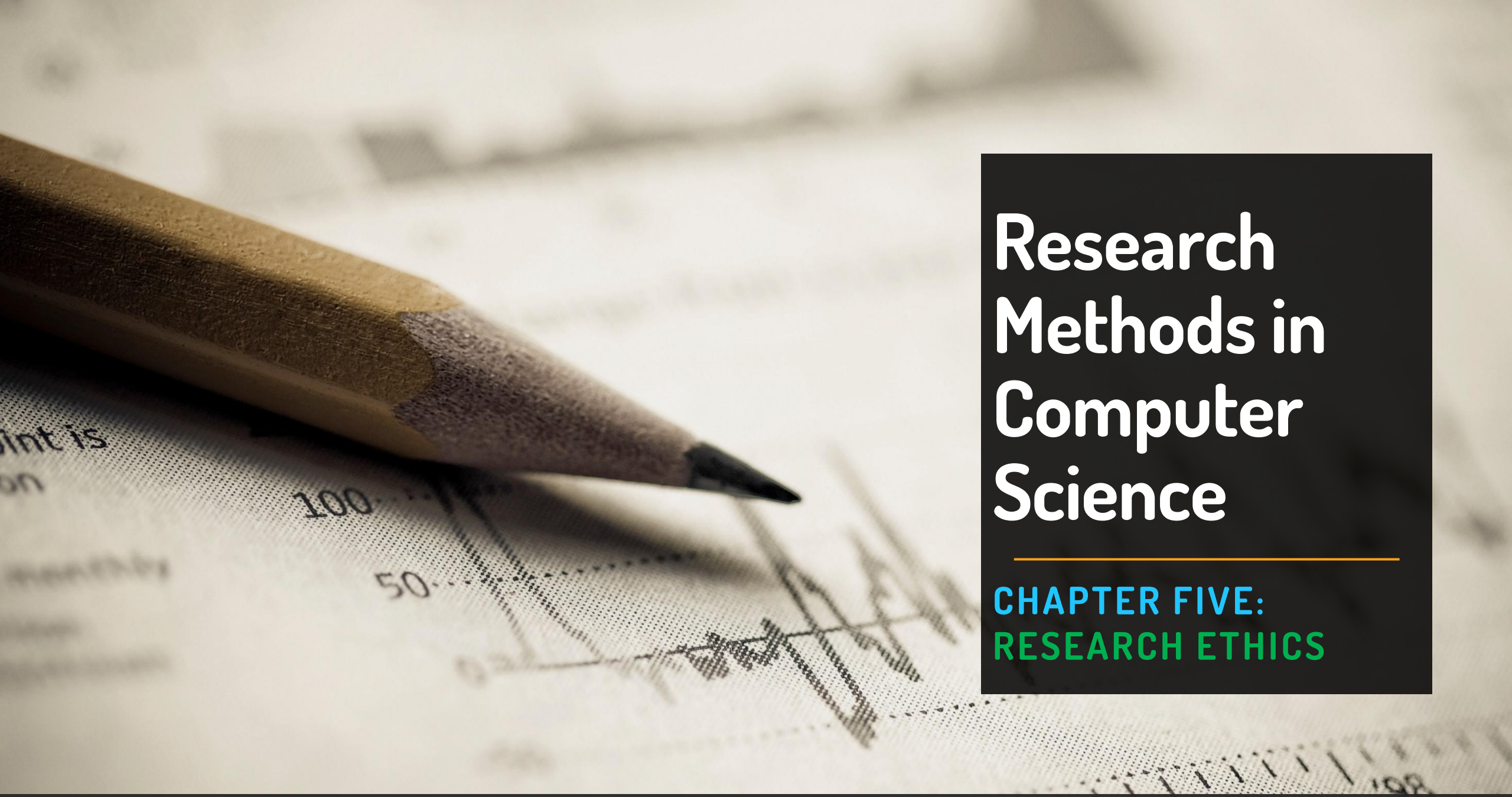

Research Methods in Computer Science

CHAPTER FIVE: RESEARCH ETHICS

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- Ethical Issues in Research
- Plagiarism, Falsification, Fabrication
- Academic Honesty Related Issues – Ex. Misleading Authorship
- Other Ethical Issues in Computing

Ethical Issues in Research

- "The most important human endeavour is the striving for **morality** in our actions. Our inner balance and even our very existence depend on it. Only morality in our actions can give beauty and dignity to life" – Albert Einstein.
- **Ethics:** Ethics is a set of standards, a code, or value system, worked out from human reason and experience, by which free human actions are determined as ultimately right or wrong, good, or evil. If acting agrees with these standards, it is ethical, otherwise, unethical.
- **Research Ethics:** Research ethics refer to some of the genres that researchers follow to protect the rights in developing research strategies and building a trusted relationship between the study participants and investigator.
- **Ethics or moral philosophy** is a branch of philosophy with standards or codes or value systems and involves defending, systematizing, recommending concepts of right, and minimizing wrong conduct.
- In a scientific study, research ethics plays a crucial role, and it refers to a persistent experience towards producing new knowledge for the sake of unveiling a new stream of ideas in academia.

Ethical Issues in Research

Basic Principles of Research Ethics (*https://rb.gy/wnwtkz*)

- Major **ethical requirements** of scientific study include **five fundamental ethical principles**, which needs to be considered at the research planning and designing phases:

  1. **Minimizing the Risk of Harm**: No research is supposed to do any harm to participants. Even then, participants could be subjected to several types of harm, such as physical harm, psychological distress, social humiliation, financial loss, and invasion into privacy by breaking the anonymity rule.

  2. **Obtaining Informed Consent**: "Informed consent" is known as the cornerstone of ethical research. The informed consent process is also known as a contract between the researcher and the participants. The term consists of two important elements, and each requires careful consideration: "**informed**" and "**consent**." The required information in the consent form should include the purpose of the research, what will be asked to them, the methods being used, how the data will be used, the possible outcomes of the research, and associated inconveniences and risks/consequences (if any) that the participants might face.

Ethical Issues in Research

Basic Principles of Research Ethics

3. **Protecting Anonymity and Confidentiality:** If the participants are willing to volunteer to share private or sensitive information, the researchers must ensure extra efforts to keep the information confidential. Insensitive data collection methods might hurt the research participants' privacy. It is likely to face a greater risk of data breach after collection if data is not stored and treated sensibly, including even during the publication process.
4. **Avoiding Deceptive Practices:** The deception of subjects may occasionally be a requirement in research studies, known as covert research. This is typically needed to promote scientific validity, where subjects are given false or incomplete information about the research to obtain unbiased data with respect to their attitudes and behaviour. This is generally used when complete or truthful disclosure is expected to produce biased results.

Ethical Issues in Research

Basic Principles of Research Ethics

5. **Providing the Right to Withdraw :** The right to withdraw, by design, is to protect research participants from an inherent uncertainty, information imbalance, inability to hedge, and untoward bodily invasion. This is important because it strengthens public trust in the research institution and the research domain. Research participants must have the right to withdraw anytime at will from the research process, except for covert observations. The research team must ensure that when a participant considers withdrawing from the research process, there should not be any way to stop them from withdrawing

Ethical Issues in Research

Fabrication, Falsification, and Plagiarism

- **Fabrication, Falsification,** and **Plagiarism** (usually referred to as **FFP**) are three ethical issues characterized as **research misconduct.**
- The first two (**Fabrication** and **falsification**) refer to **misdemeanours** during research, and the third (**plagiarism**) refers to **misconduct** during the publication or dissemination process.
- All three are considered in one category because research is incomplete unless it is disseminated.
- **Fabrication** and **Falsification** of data amount to **fraud**, and will result in some of the researchers, who read this concocted research, wasting their time if they try to follow up on it.

Ethical Issues in Research

Fabrication, Falsification, and Plagiarism

- **Plagiarism** is a more commonly discussed misdemeanour because there is a perpetrator of the misdemeanour, and there is also a clearly identifiable victim in the researcher whose work is being plagiarized and whose credit is being stolen.
- **Fabrication** and **falsification** are **fraud**, while **plagiarism** is **theft** of the **victim's credit.** Such unethical behaviour is different from the behaviour involving the living subjects of a research process, encountered in life-sciences or social-sciences.
- **Fabrication** and **falsification** refer to **not being totally honest in trying to find answers**. Researchers may, for reasons that are guessed to be attempts to get recognition or rewards, not be honest while conducting the experimental or theoretical investigation central to the research, or may conduct the investigation honestly but be unethical during its analyses.

Ethical Issues in Research

Fabrication, Falsification, and Plagiarism

- Plagiarism is when the researchers were honest in both these stages of the research project, but became unethical while writing their research output.
- **Plagiarism** is defined as "the appropriation of another person's ideas, processes, results, or words without giving appropriate credit".
- "**Plagiarism**" means the practice of taking someone else's work or idea and passing them as one's own."
- We can refer to the use of another person's words as **"text-plagiarism",** and the use of another person's ideas as **"idea-plagiarism".**
- **Text-plagiarism** is also referred to as **"cut-and-paste plagiarism",** and is easily detected by commercially available software that detects text-similarity.

Ethical Issues in Research

Fabrication, Falsification, and Plagiarism

Self-plagiarism: occurs when you **reuse your own previously published work without proper citation or disclosure**, in a way that misrepresents it as new/original content. Key forms include:

1. **Duplicate/Overlapping Publication**: Publishing the same data, findings, or substantial text in multiple venues without informing editors/readers.
2. **Salami Slicing**: Breaking a single study into multiple small papers to artificially increase publication count.
3. **Recycling Text Without Attribution**: Copying substantial portions from your previous papers without quoting or citing yourself.

Ethical Issues in Research

Academic Honesty Related Issues – Ex. Misleading Authorship

- **Academic dishonesty** is defined as the intentional engagement in prohibited behaviours, such as cheating on exams, copying assignments, and plagiarism, in order to gain an unfair advantage in an academic setting.
- The **issue of academic honesty** is a sensitive one for a university because it is so central to the individual
  - learner's self-identity, the campus's academic mission, the university's reputation, and the qualifications it confers.
- While universities strive to build learning cultures that support honest research and teaching, academic integrity goes beyond the quality of work to the moral fiber of each generation of learners, and these values include **"honesty, trust, fairness, respect,** and **responsibility"**

Ethical Issues in Research

- **Academic integrity** is the value of **being honest, ethical,** and **thorough** in your academic work. It allows readers to trust that you aren't misrepresenting your findings or taking credit for the work of others.

Types of academic dishonesty

- **Plagiarism:** Copying someone else's work and passing it off as your own, without giving proper credit. (**E.g.** Copying and pasting parts of a source you found online without citing it)
- **Cheating:** Using unauthorized sources or devices to help you achieve an outcome you wouldn't have on your own. (**E.g.** Copying someone's answers on an exam)
- **Contract cheating:** Paying or bribing someone to help you cheat.
- **Facilitation of academic dishonesty**
- **Data fabrication, Collusion and Deceit**

Ethical Issues in Research

Why does academic integrity matter?

- There are various reasons you might be tempted to resort to academic dishonesty: pressure to achieve, time management struggles, or difficulty with a course. But academic dishonesty hurts you, your peers, and the learning process. It's:
  - **Unfair to the plagiarized author**
  - **Unfair to other students who did not cheat**
  - **Damaging to your own learning**
  - **Harmful if published research contains misleading information**
  - **Dangerous if you don't properly learn the fundamentals in some contexts (e.g., lab work)**

Ethical Issues in Research

Causes of Academic Dishonesty

- Literature on academic dishonesty cites a number of factors that contribute to dishonest academic practices (*Whitley and Keith-Spiegel, 2002*). Contributing factors include:
  - **Peer pressure**
  - **Performance anxiety**
  - **Excuse making**
  - **Inability to manage the demands of student life**
  - **Situations that encourage academic dishonesty**
  - **Self-justification habits**
  - **Unfamiliarity with what constitutes academic dishonesty**
  - **Lack of understanding about consequences**

Ethical Issues in Research

Examples of academic dishonesty

Mild academic dishonesty:

- Faking illness to skip a class,
- Asking for a classmate's notes from a special review session held by your professor that you did not attend,
- Crowdsourcing or collaborating with others on a homework assignment,
- Citing a source you didn't actually read in a paper.

Moderate academic dishonesty:

- Cheating on a pop quiz
- Peeking at your notes on a take-home exam that was supposed to be closed-book
- Resubmitting a paper that you had already submitted for a different course (self-plagiarism)
- Forging a doctor's note to get an extension on an assignment

Ethical Issues in Research

Examples of academic dishonesty

Severe academic dishonesty:

- Fabricating experimental results or data to prove your hypothesis in a lab environment
- Buying a pre-written essay online or answers to a test
- Falsifying a family emergency to get out of taking a final exam
- Taking a test for a friend

Ethical Issues in Research

Other Ethical Issues in Computing

https://online.keele.ac.uk/ethical-challenges-in-computer-science/

- Ensuring ethics is fully integrated into the field of computer science – from education to real-world practice – serves to develop tech that benefits humanity while minimising harm. The absence of ethics and ethical standards can, quite simply, have the reverse effect.
- When it comes to technological advancements and practices, any inappropriate or unethical treatment of individuals or communities falls under intense scrutiny.
- As such, tech giants and other businesses– from Microsoft to OpenAI – continually strive for integrity in their practices, from ethical decision-making to day-to-day operations. Failure to do so could result in business failure – as well as other far more serious issues.

Ethical Issues in Research

Other Ethical Issues in Computing

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What are some of the ethical issues in computer science and technology?

There are a host of ethical implications and ethical problems posed by computer science, automation, and technology. **Major ethical issues in computer science:**

- **AI Ethics: Use of artificial intelligence (Use of autonomous technologies)**
- **Negative environmental impact**
- **Misuse of personal and sensitive data (Data privacy and security)**
- **Deep Fakes and misinformation**
- **Intellectual property rights**
- **Cyber bullying**

- **For more: https://rb.gy/ek8au0**

Ethical Issues in Research

Other Ethical Issues in Computing

Best practices in computer ethics?

- Being accountable
- Respecting privacy
- Investing in continuous learning and development
- Acting fairly and equitably
- Promoting honesty and transparency